

HAVANA JOSE MARTI, Cuba

WMO: 782240

Lat: 22.989N

Lon: 82.409W

Elev: 64

StdP: 100.56

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	12.1	7.8	6.6	14.5	9.0	7.2	15.6	10.3	21.7	9.3	21.6	0.7	230	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	33.1	24.9	32.8	24.9	32.1	24.8	27.5	30.3	26.8	30.0	26.2	29.7	3.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.8	29.4	26.1	21.6	28.6	25.2	20.5	27.9	88.1	30.0	84.8	30.5	82.0	29.5	34.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.2	6.9	34.3	1.7	0.6	5.7	34.8	4.7	35.2	3.8	35.5	2.6	36.0		
(5)			6.6	27.8	1.5	1.8	5.4	29.1	4.5	30.1	3.7	31.1	2.6	32.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	20.7	21.7	22.5	24.4	26.1	27.1	27.4	27.5	27.0	25.7	23.0	21.9	(6)	(7)
	DBStd	3.11	2.85	2.71	2.51	2.07	1.38	1.06	0.82	0.81	0.87	1.68	2.34	2.65	(7)	(8)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	(9)
	HDD18.3	27	12	6	2	0	0	0	0	0	0	0	1	6	(9)	(10)
	CDD10.0	5327	332	327	388	432	498	512	540	542	511	488	389	370	(10)	(11)
	CDD18.3	2313	86	99	132	182	239	262	282	284	261	229	140	118	(11)	(12)
	CDH23.3	19395	509	704	1056	1552	2126	2412	2665	2630	2242	1831	955	714	(12)	(13)
	CDH26.7	6856	84	161	291	534	801	926	1126	1113	867	595	208	151	(13)	(14)
(14) Wind	WSAvg	3.1	3.5	3.4	3.8	3.7	3.3	2.8	2.6	2.5	2.5	3.1	3.3	3.2	(14)	(15)
(15) Precipitation	PrecAvg	1213	46	39	40	77	130	185	134	157	171	131	56	47	(15)	(16)
	PrecMax	1532	172	123	98	149	399	403	212	287	331	365	153	163	(16)	(17)
	PrecMin	918	4	7	18	24	39	81	62	95	77	37	2	7	(17)	(18)
	PrecStd	160	40	30	20	32	73	75	35	41	52	74	37	36	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	30.2	31.2	32.2	33.2	33.8	33.8	33.9	33.1	32.3	31.1	30.2	(19)	(20)
		MCWB	21.5	21.6	22.1	22.4	23.7	25.1	24.9	25.1	25.1	24.8	23.8	23.0	(20)	(21)
	2%	DB	28.8	29.8	30.2	31.7	32.2	32.8	33.0	33.1	32.2	31.2	30.0	29.2	(21)	(22)
		MCWB	21.6	21.6	22.4	22.6	23.8	25.1	24.9	25.2	25.3	25.0	23.6	22.6	(22)	(23)
	5%	DB	27.8	28.8	29.2	30.9	31.5	32.0	32.2	32.2	31.9	30.9	29.0	28.2	(23)	(24)
		MCWB	21.4	21.6	21.9	22.5	23.7	25.0	25.0	25.3	25.3	25.0	23.0	22.2	(24)	(25)
	10%	DB	26.4	27.8	28.4	29.9	30.8	31.1	31.8	31.9	31.0	30.0	27.9	27.2	(25)	(26)
		MCWB	20.9	21.4	21.4	22.3	23.6	25.0	25.1	25.3	25.3	24.7	22.4	21.7	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	24.0	25.1	25.2	26.2	27.6	27.8	28.4	28.1	27.9	25.5	24.7	(27)	(28)
		MCDB	26.7	27.5	28.8	29.4	30.2	30.7	30.7	31.2	30.3	30.0	28.3	27.8	(28)	(29)
	2%	WB	23.1	23.2	24.0	24.4	25.6	26.7	26.8	27.4	27.1	26.6	24.6	23.8	(29)	(30)
		MCDB	26.2	26.8	27.7	28.7	29.8	30.1	30.3	30.4	29.9	29.1	27.8	27.2	(30)	(31)
	5%	WB	22.4	22.6	23.3	23.8	25.0	26.2	26.2	26.5	26.2	25.9	24.0	23.2	(31)	(32)
		MCDB	25.7	26.3	27.1	28.0	29.4	29.7	30.0	30.1	29.5	28.7	27.2	26.5	(32)	(33)
	10%	WB	21.7	22.1	22.5	23.2	24.4	25.7	25.7	25.9	25.8	25.2	23.5	22.6	(33)	(34)
		MCDB	25.0	25.9	26.5	27.6	28.7	29.3	29.8	29.9	29.3	28.4	26.5	25.8	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	11.3	11.1	9.7	8.8	9.6	9.5	8.9	8.7	9.8	10.1	(35)	(36)
		MCDBR	11.2	11.9	11.8	11.6	10.6	9.7	10.1	10.2	9.8	9.6	10.2	10.9	(36)	(37)
		MCWBR	5.5	5.4	5.2	4.5	4.1	4.1	4.2	4.3	4.2	4.2	4.8	5.3	(37)	(38)
	5% WB	MCDBR	9.8	10.5	10.5	10.3	9.3	8.9	9.6	9.5	8.8	8.3	8.5	9.3	(38)	(39)
		MCWBR	5.4	5.2	5.2	4.6	4.2	4.3	4.4	4.7	4.4	4.2	4.5	4.9	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.359	0.367	0.375	0.400	0.426	0.447	0.455	0.431	0.415	0.404	0.375	0.370	(40)	(41)
	taud		2.524	2.507	2.469	2.402	2.371	2.361	2.329	2.416	2.454	2.467	2.497	2.513	(41)	(42)
	Ebn at Noon		908	922	927	903	872	846	840	863	875	873	882	881	(42)	(43)
	Edh at Noon		95	103	112	121	125	125	129	118	112	106	97	94	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		3.97	4.82	5.84	6.52	6.39	5.92	6.43	6.14	5.41	4.70	4.10	3.76	(44)	(45)
	RadStd		0.31	0.32	0.22	0.22	0.61	0.43	0.30	0.21	0.29	0.34	0.20	0.29	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	+0.89	N/A	N/A	N/A	N/A	N/A	-11	N/A	N/A

Nomenclature: See separate page